

In this problem set you will train and evaluate a classification algorithm to determine whether or not a fine needle aspiration biopsy is cancerous (malignant) or non-cancerous (benign). The data were downloaded from the UC Irvine Machine Learning Repository and lightly processed. Here is a brief glimpse at some of the columns. Use this glimpse to answer the following questions.

```
# A tibble: 7 × 7
  diagnosis radius_mean texture_mean area_mean radius_sd texture_sd area_sd
  <fct>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>
1 M          16.1        20.7        799.        0.569        1.07        54.2
2 M          19.8        22.2       1260         0.758        1.02       112.
3 B          13.5        14.4        566.        0.270        0.789        23.6
4 B          13.1        15.7        520         0.185        0.748        14.7
5 B           9.50        12.4        274.        0.277        0.977        15.7
6 M          15.3        14.3        704.        0.439        0.710        44.9
7 M          21.2        23.0       1404         0.692        1.13       94.0
```

Question 1

What is the unit of observation in this data frame?

Question 2

We will be fitting models to output a diagnosis (“benign” or “malignant”). This is a categorical outcome. Which level will be considered the reference level by default in R and why?

Question 3

If you were to deploy your method in a clinical setting to help diagnose cancer, would it be worse to misclassify a benign case or to misclassify a malignant case? Explain your rationale in at least two sentences.

Question 4

Based on the glimpse, use a plot to compare the `radius_mean` for benign vs. malignant biopsies, *side-by-side*. Make sure to give your label your axes and give your plot a title. Give a shape which matches **your** expectation of the phenomenon and **explain** your choice in at least one sentence.

Question 5

Based on your previous sketch, what biopsies are you prepared to classify as malignant versus benign? Fill in the blanks below to make a decision rule.

```
If radius_mean > _____: predict _____  
Otherwise predict _____
```

Question 6

Based on the glimpse, sketch a plot that examines the association between two predictors, `radius_mean` and `area_mean`. Make sure to give your label your axes and give your plot a title. Give a shape which matches **your** expectation of the phenomenon and **explain** your choice in at least one sentence.

Question 7

In many realms of medicine, classification algorithms can be more accurate than the most well-trained medical doctors. What is gained and what is lost by shifting to algorithmic diagnoses? Although a book could be written about this topic, please answer in one paragraph.